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Law domain, AI chatbot, LLMD4LH,

This gap motivates the development of our proposed AI assistant, which aims to provide grounded, explainable, and robust answers based directly on formal Slovenian legal sources while maintaining accessibility for general users.

Corpus Analysis

For the development of our system, we selected *COLESLAW 1.0*, a recent collection of Slovenian legal texts published in the CLARIN.SI repository. [3] The corpus contains 547,799 texts and 771,931,725 words distributed across 12 JSONL files. At the highest level, it combines four complementary legal domains: the PISRS legal information system, the *SodnaPraksa* collection of court decisions, the Constitutional Court collection (*USRS*), and texts from the *Uradni list*. This is useful for our task because the dataset does not focus only on laws or only on court decisions, but instead combines several types of legal texts in one place.

A closer look at the internal structure shows that the corpus is not balanced across domains, which is important for downstream modeling choices. The largest share by number of documents comes from *Uradni list*, which contributes 274,531 texts, or just over half of the corpus, while *SodnaPraksa* contributes 217,409 texts. In terms of word count, however, *SodnaPraksa* becomes the dominant source with 352.3 million words (45.6% of all words), followed by *Uradni list* with 296.7 million words (38.4%). PISRS is much smaller in the number of documents, with 40,749 texts, but still contributes 96.8 million words, which indicates that its documents are on average longer and denser than many gazette entries. The USRS collection adds a further 15,110 constitutional court documents and 26.2 million words.

The corpus also contains noticeable differences between subcollections. For example, *sp_courts* is the largest individual part of the dataset, while *ul-razglasni* contains shorter announcement-style texts. Some legislative collections in PISRS are much longer, especially the register of regulations and draft regulations. Because of this, the corpus includes both short notices and long legal documents. In our opinion, this makes COLESLAW 1.0 a good starting point for building a Slovenian legal assistant, since it is large, diverse, and based on official public sources.

Initial Ideas

At this stage, our main idea is to base the system on retrieval-augmented generation. Instead of letting the model answer only from its internal knowledge, we would first retrieve relevant passages from official Slovenian legal documents and then generate the answer from those passages. This should help improve accuracy and also make it easier to show the user where the answer comes from.

To reduce hallucinations, we would like the assistant to stay close to the retrieved documents, include references to the source texts, and clearly say when the available information is not enough for a reliable answer. We also expect that careful chunking of long documents, filtering by source type, and reranking of retrieved passages could further improve the quality of the final responses.

References

- [1]
- [2] MojMaliPravnik.net. Mojmalipravnik.net - pravni nasveti za majhne in velike.
- [3] Miha Malenšek, Slavko Žitnik, Aleš Završnik, Saša Krajnc, Primož Križnar, and Marko Bajec. Collection of slovenian legal texts coleslaw 1.0, 2024. CLARIN.SI repository.